Rethinking Fronthaul Topologies for Cell-Free 6G Networks

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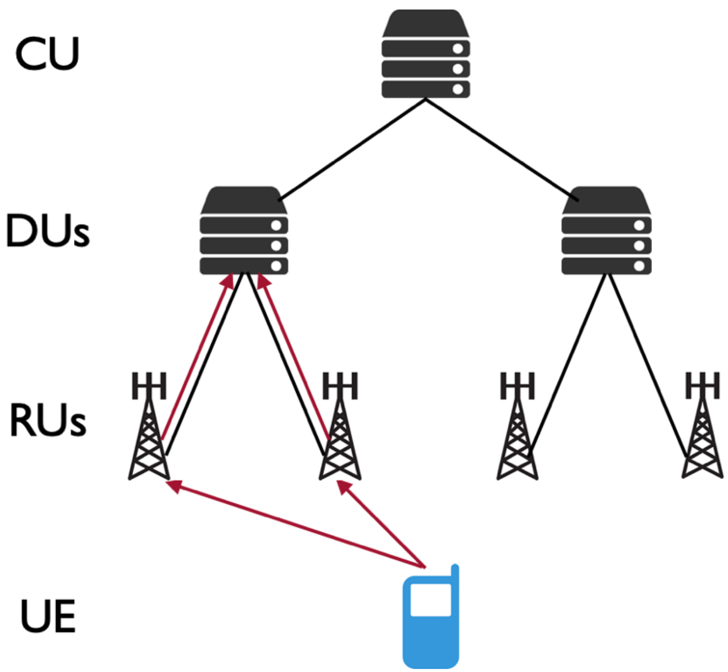
What this presentation will cover

- How the cell-free paradigm will impact current fronthaul network topologies
- The network model we used
- The implications on link capacities
- What alternative topologies could be considered
- How these topologies perform in terms of bandwidth
- How DU assignment algorithms influence the bandwidth requirements

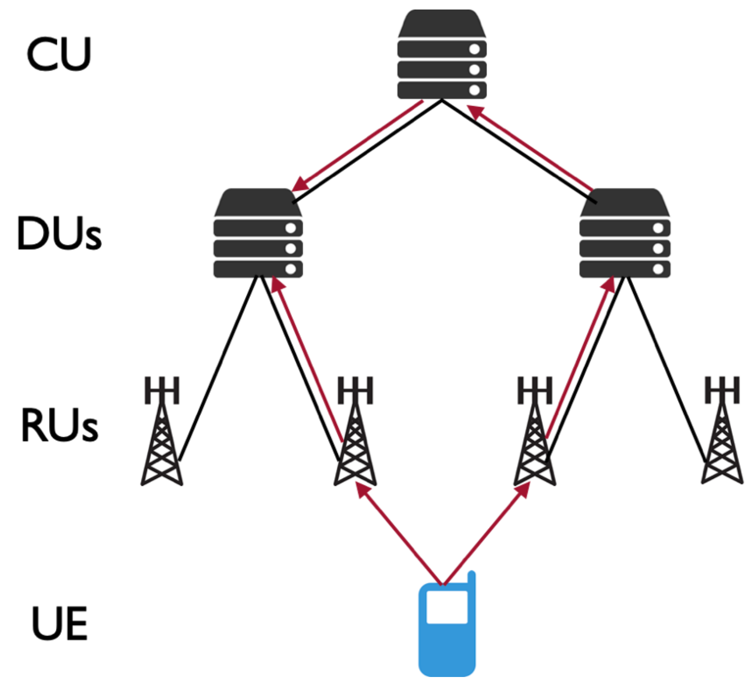
What this presentation will NOT cover

- Discussions about the physical layer model
- If you are interested, it's described in detail in our paper and in previous work:
    - Li, Z., Göttsch, F., Li, S., Chen, M., & Caire, G. (2024). Joint fronthaul load balancing and computation resource allocation in cell-free user-centric massive MIMO networks. *IEEE Transactions on Wireless Communications*, *23*(10), 14125-14139.
    - Göttsch, F., Franke, M., Pourdamghani, A., Caire, G., & Schmid, S. (2025). Optimizing Fronthaul Quantization for Flexible User Load in Cell-Free Massive MIMO. *IEEE Globecom Workshop on Resilience in Next-Generation Wireless Communication Networks*

Cell-Free vs. Cellular: A Paradigm Shift

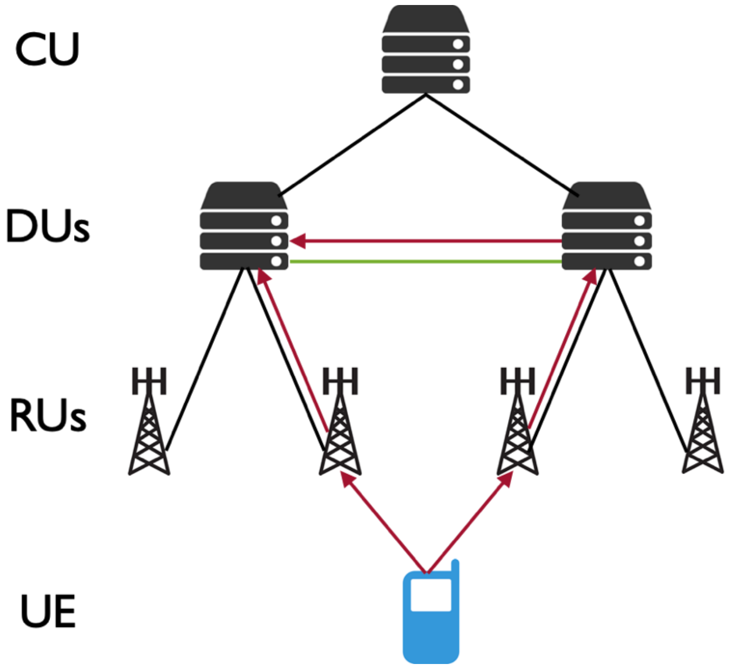


Distributed MIMO

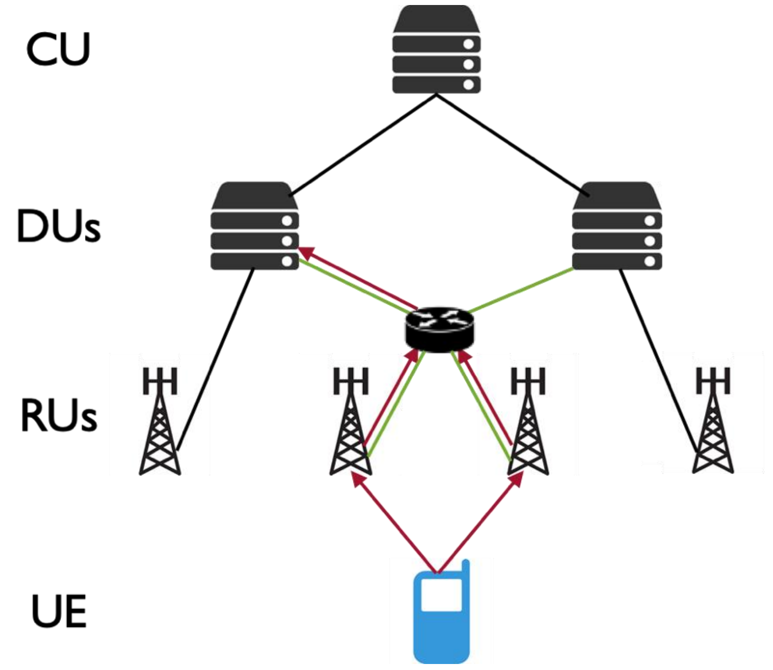


Cell-Free

The issue with trees



Inter-DU

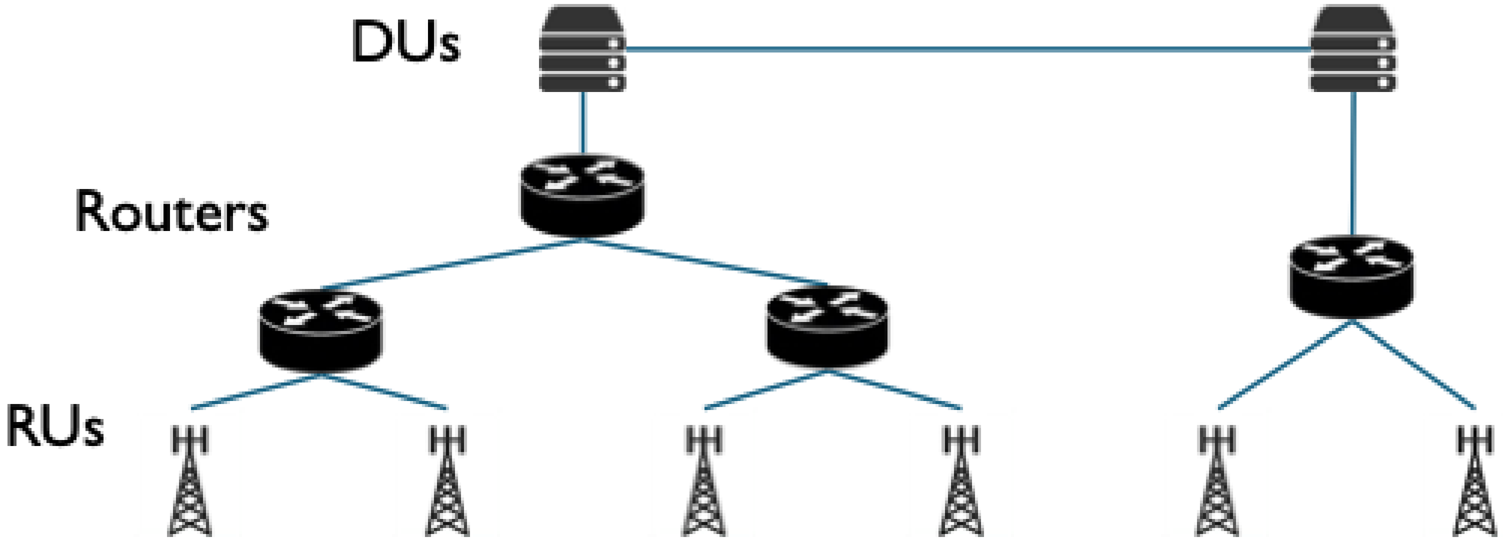


Routed FH

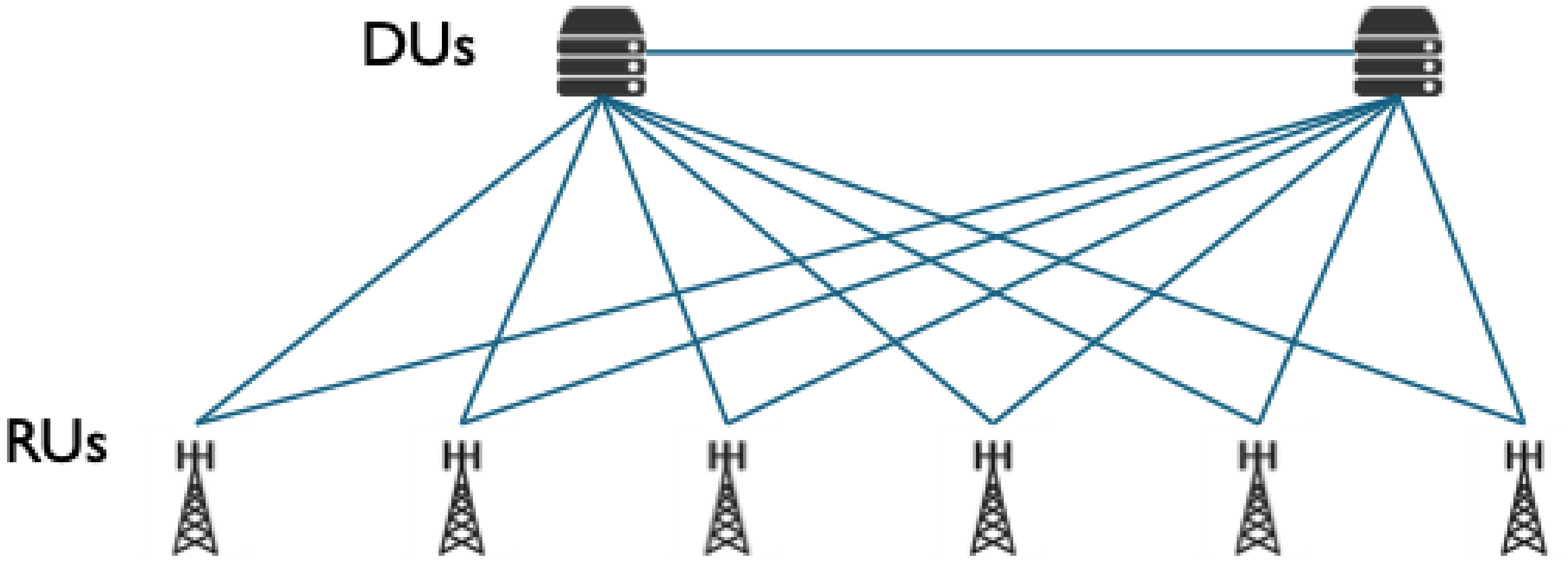
Network model

- Abstracted away concrete delay/latency
- Too complex to calculate (especially queueing delay) and depends on technology
  - I.e.: Optical switching might have very low latency beyond pure speed of light delay
- Instead using a maximum hop count constraint
- Each DU has a maximum number of UEs it can support
- We calculate fronthaul demand for each UE-RU pair based on our PHY layer model
- We assign a DU per UE that is handling the processing for that UE
  - This means that all traffic from/to the UE must be routed there
  - More on how to assign later
- Shortest path routing

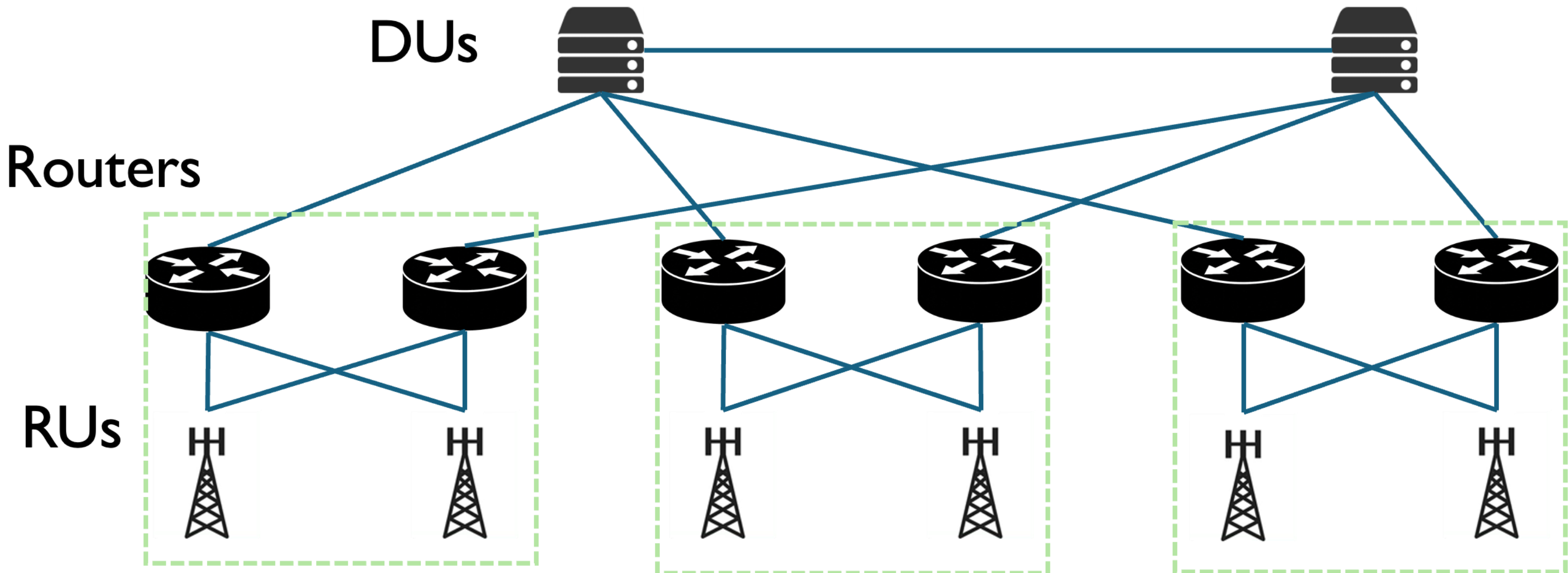
Candidate Topologies - Tree



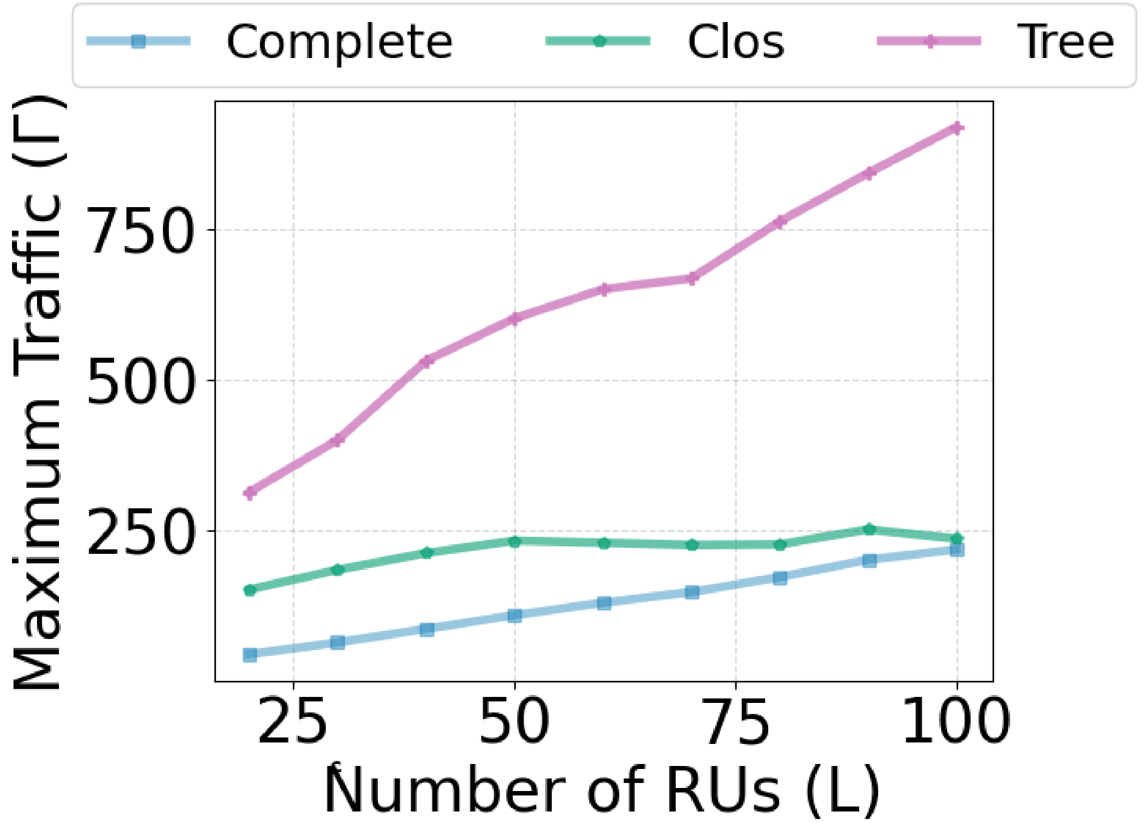
Candidate Topologies - Complete



Candidate Topologies - Clos

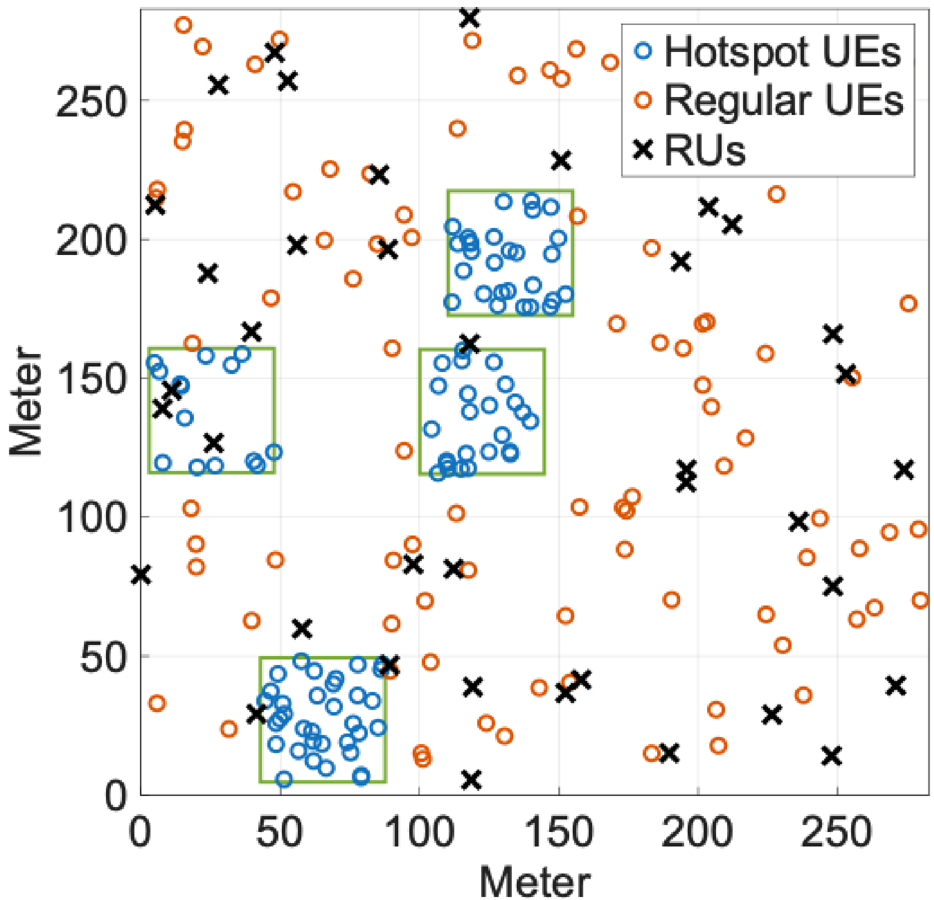


Results – Trees don't scale

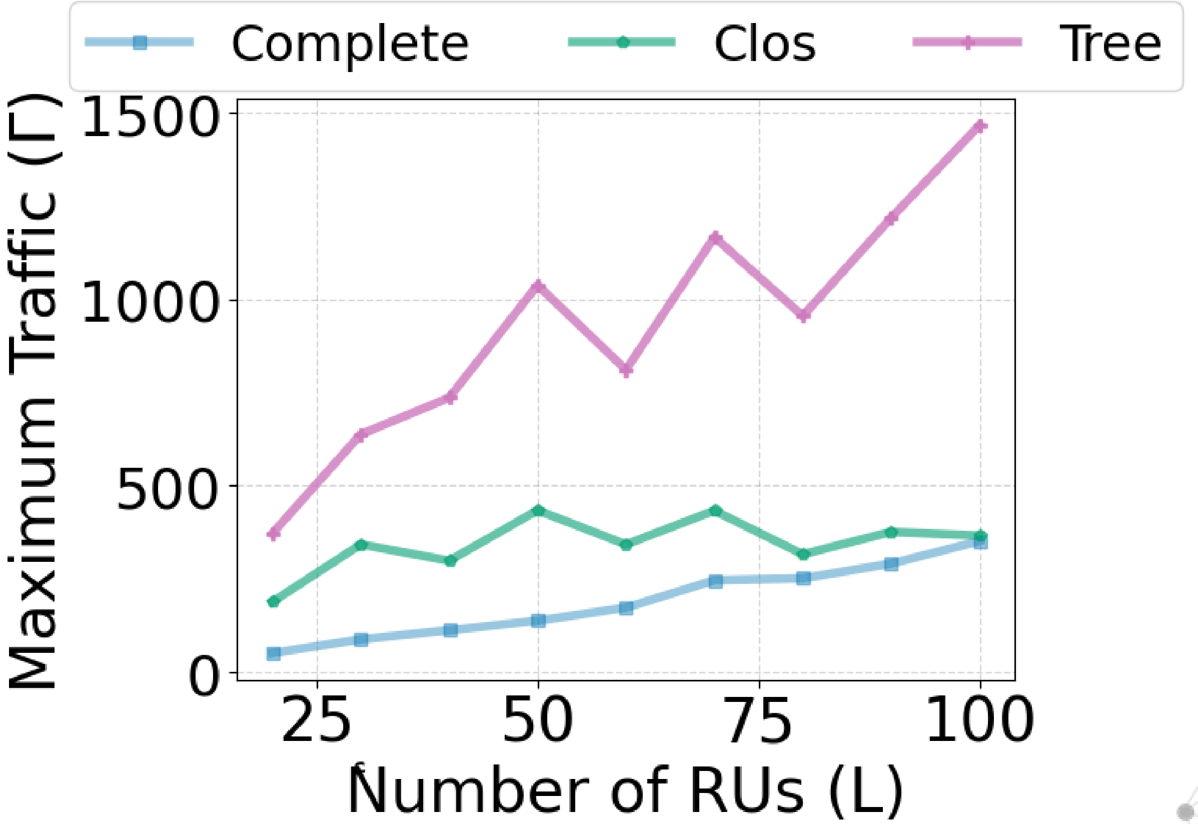


Γ = Gbps

Does Distribution of UEs matter?



Results – Topology conclusion remains

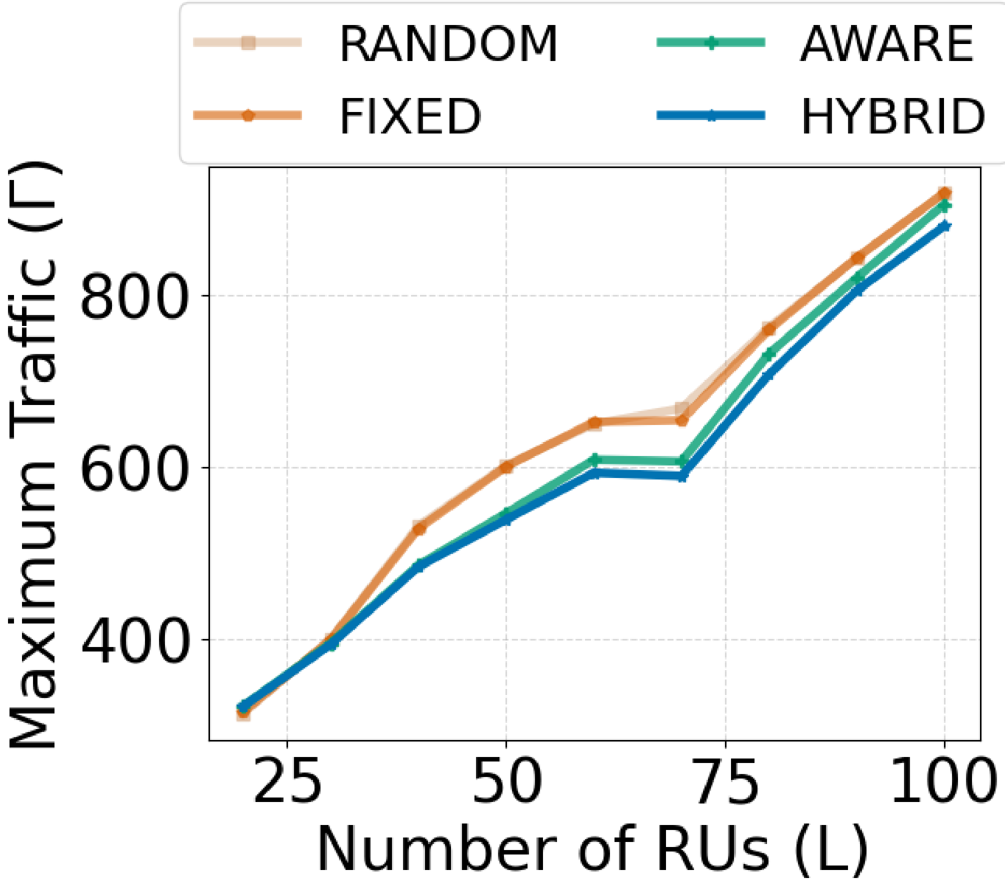


Γ = Gbps

Placement Algorithms

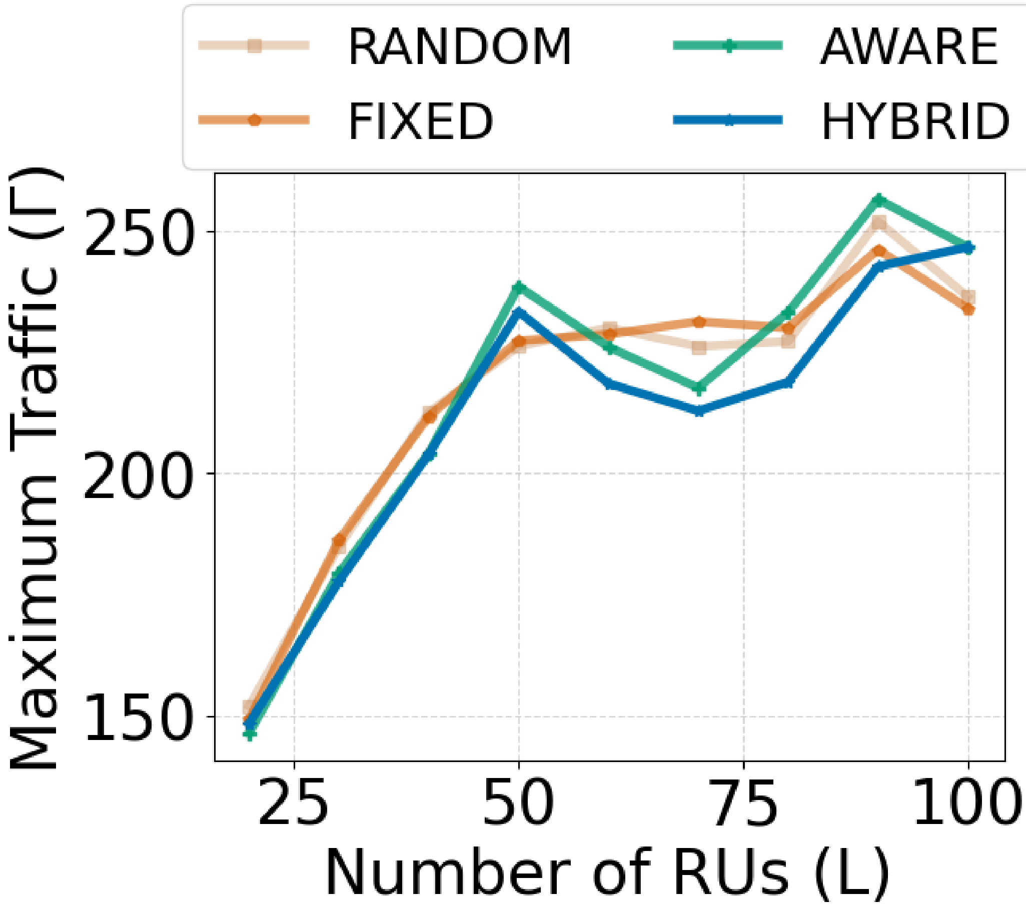
- We investigated four different algorithms for assigning DUs to UEs
- **Random:** Assigns UEs uniformly at random
- **Fixed:** Evenly divides UEs among DUs; goal is for scenarios with equal UE load distribution
- **Aware:** Compute shortest paths first, then sort by demands
- **Hybrid:** Extension of Aware algorithm to include current link utilizations

Results – Better algorithms can make trees slightly better



Γ = Gbps

Results – Other algorithms work better on Clos



Γ = Gbps

Do Placement Algorithms matter?

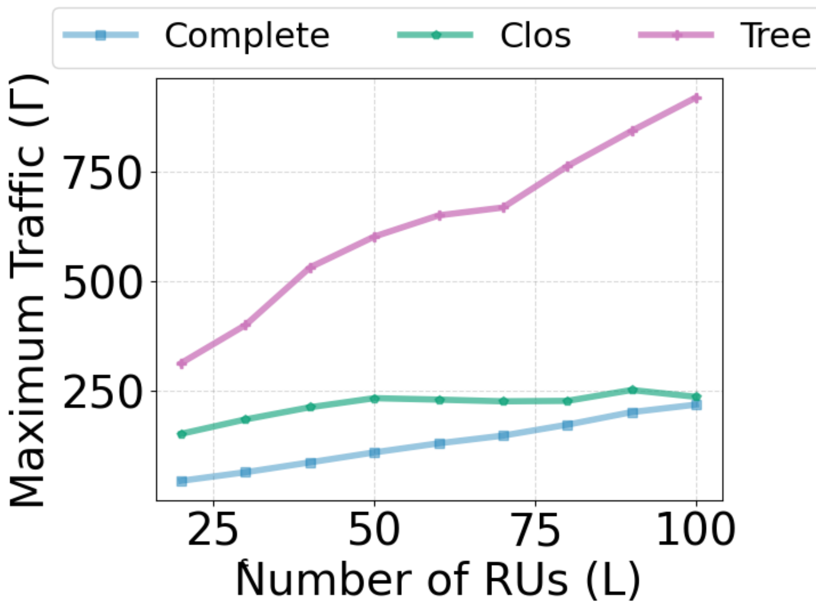
- They do, reducing maximum bandwidth by up to 20%
- Some algorithms work better on trees, some better on Clos
- However, the ones we proposed aren't anywhere close to optimal yet

Take Aways and Outlook

- Current fronthaul topology design will likely become a bottleneck in cell-free deployments
- This issue will only be compounded even further if higher frequencies like sub-THz are used
- There will likely be a need for routing in the fronthaul, adding complexity and latency
- Clos topologies, which are well-known and widely used in data-center networks, offer an attractive candidate that balances deployment complexity and efficiency
- Future work could study RAN specific topologies and better placement algorithms

Questions?

Paper



Scholar